

Apoorv Parikh - Chemical Engineer

Calgary, AB

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Professional Summary

PhD researcher specializing in **solid sorbent** for **CO₂ capture** with 3+ years of industry-funded R&D experience. Focused on advancing TRL of zeolite-based solid sorbents under industrially relevant conditions, translating lab-scale performance into scale-up and feasibility insights. Contributed to peer-reviewed publications, patent filings, and projects funding applications, with demonstrated improvements of nearly 30% in CO₂ uptake and multi-cycle stability. Keen on catalyst/adsorbent kinetics and shaping of the sorbent material.

Education

Doctor of Philosophy (PhD), Chemical and Petroleum Engineering Expected Sept 2026
University of Calgary, Canada

Supervisor: Prof. Nader Mahinpey

Thesis: Development of novel solid sorbent, geopolymer zeolite composite, for CO₂ capture from cogeneration power plants

Master of Science (Eng.), Chemical Process Technology 2019–2021
Eindhoven University of Technology (TU/e), Netherlands

Supervisor: Prof. Martin van Sint Annaland

Thesis: Investigated catalyst performance trade-offs to optimize benzene selectivity in methane dehydroaromatization

Bachelor of Engineering, Chemical Engineering 2014–2018
Gujarat Technological University, India

Industry Experience

Technology Advisor (Internship) Sept 2025–Dec 2025
Carbanaut Engineering, Calgary, AB

- Screened and down-selected organic linker chemistries for COF-based sorbents based on compatibility, stability, and synthesis constraints relevant to scale-up.
- Coordinated external adsorption and characterization testing to close data gaps required for material benchmarking.
- Synthesized experimental results into technical recommendations used to guide subsequent material development priorities.

Project Leader — Catalyst Optimization Apr 2021–Sept 2021
Avantium Chemicals, Amsterdam, Netherlands

- Designed and executed a 16-condition high-throughput screening campaign for selective hydrogenation catalysts using the Flowrence[®] platform.
- Analyzed conversion–selectivity trade-offs across catalyst formulations and dilution ratios to identify optimal operating windows.
- Delivered experimentally supported recommendations that informed catalyst selection and subsequent process development.

Research Experience

CO₂ Capture Research & Development (PhD Research) University of Calgary

2021–Present

- Developed Zeolite 13X-based and geopolymer composite sorbents targeting low-temperature CO₂ capture under cogeneration-relevant conditions.
 - Increased CO₂ uptake from 3.2 to 4.7 mmol g⁻¹ (nearly 30% improvement), with performance retained over 20 adsorption–desorption cycles.
 - Used statistically designed experiments to identify synthesis parameters with the strongest impact on adsorption capacity and stability.
 - Generated kinetic, equilibrium, and breakthrough data used to benchmark materials against commercial sorbents.
 - Built Aspen Adsorption models to assess feasibility and sensitivity of adsorption-based capture concepts, informing scale-up assumptions.
 - Contributed to successful COSIA- and NSERC-funded proposals and technical presentations, supporting over \$400k in awarded research funding.
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Publications, Patents & Conferences

- **Parikh, A.**; Mahinpey, N. *Optimizing Zeolite 13X Synthesis from Kaolin Clay for Higher CO₂ Adsorption: A Statistical Approach*. *ChemCatChem* **2025**, 17, e00980.
<https://doi.org/10.1002/cctc.202500980>.
 - **Parikh, A.**; Mahinpey, N. *GHGT-17 Conference Proceedings* **2024**.
<https://doi.org/10.2139/ssrn.5067692>.
 - **Parikh, A.** First-named inventor. *In-situ Synthesis of Zeolite 13X–Geopolymer Composite Sorbents for Low-Temperature CO₂ Capture*. U.S. Provisional Patent Application No. 63/931,682.
 - **Parikh, A.** Contributing author. *Chemical Looping Combustion*, Elsevier **2026**.
<https://doi.org/10.1016/B978-0-443-26659-1.00001-7>;
<https://doi.org/10.1016/B978-0-443-26659-1.00004-2>.
 - **Parikh, A.** Presenter at CScHE 2023, CScHE 2024, and GHGT-17.
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Technical Skills

Programming & Data Analysis: Python (NumPy, pandas, SciPy, matplotlib, scikit-learn)

Modeling & Simulation: Aspen Plus, MATLAB

Experimental & Characterization: TGA, fixed-bed adsorption columns, adsorption isotherms, XRD, SEM

Methods: Design of Experiments (DOE), kinetic analysis, statistical modeling, process modeling, techno-economic assessment

Teaching Experience

- Delivered lectures and coordinated industry field visits for CCUS courses at the University of Calgary.
- Instructed 45+ mid-career professionals in techno-economic analysis and CCUS feasibility planning at NAIT.